

Controlled q -Voter Theory for Round-Level Feedback

A Step-by-Step Tutorial on Exact Transfer Entropy, Finite-Horizon Currents, Variance, Fano Statistics, and SNR²

Based on the five-page theory note dated 14 August 2026

Purpose. This document is intentionally pedagogical. Every symbol is defined, the probability calculations are unpacked, algebraic simplifications are shown explicitly, and the logic of each construction is explained before the formal result is used. The original transfer-entropy derivation is preserved and expanded with a second, complementary family of observables: finite-horizon population currents and their fluctuations.

Abstract

This tutorial derives, step by step, the controlled unanimity q -voter model with round-level feedback and the transfer-entropy quantities reported in the source theory note, and then extends the same exact finite-population machinery to signed population currents and their fluctuations. The population consists of N binary agents. At the start of each round, a controller samples q_c agents, chooses either NoOp or Advocate through a soft sigmoid policy, and, if it advocates target opinion A , replaces one of the q social-input slots at exactly b of the N microscopic update positions. We first derive the ordinary and controlled one-update kernels K_0 and K_1 , then construct the exact whole-round kernels R_0 and R_1 , including the dynamic-programming recursion that averages exactly over all uniformly preallocated control schedules. We then derive the exact finite- N local transfer entropy $T(n) = I(U_k; N_{k+1} \mid N_k = n)$ and show why it is a weighted Jensen–Shannon divergence. After the original transfer-entropy development, we introduce the finite-horizon target current $J^{(K)} = N_K - N_0$, derive its exact probability distribution from powers of the same round kernels, and obtain exact expressions for its mean, variance, Fano-like dispersion, inverse-Fano current precision, and squared signal-to-noise ratio. For $q = 1$ we derive the scalar closed-form one-round mean current and show why it agrees with the exact kernel calculation. Throughout, explanatory figures and worked examples connect the formulas to their probabilistic meaning and to the quantities that can be estimated from repeated multi-agent simulations.

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1 What problem are we solving?

We study a population of N agents. Every agent currently supports one of two opinions,

$$A \quad \text{or} \quad B.$$

The population evolves through a unanimity q -voter rule: a focal agent switches only when all q social inputs unanimously support the opposite opinion.

An external controller tries to favor opinion A . The controller does not directly overwrite the focal agent's decision rule. Instead, it can insert one pro- A social input into selected microscopic updates. Its action is chosen only once per population round after observing a finite sample of the population.

The central information-theoretic question is

$$I(U_k; N_{k+1} \mid N_k), \quad (1)$$

where U_k is the controller's round-level action, N_k is the number of A supporters at the beginning of round k , and N_{k+1} is the number at the beginning of the next round.

A second, complementary question is not about information but about *directed population motion*:

$$J^{(K)} = N_K - N_0. \quad (2)$$

This asks how many net supporters the target gained or lost over a complete finite episode. Repeating the same experiment produces a distribution of $J^{(K)}$, from which we can measure both average steering and its reliability.

Intuition

Transfer entropy and current answer different questions. Transfer entropy asks whether variation in the controller action makes the next population state more predictable. Current asks whether the population actually moves toward or away from the target. A controller that always advocates can generate a large positive current even though U_k has no variation and therefore its action-to-population transfer entropy is zero by construction. This is why the two observable families are complementary rather than redundant.

Intuition

Conditioning on N_k asks a very precise causal-predictive question: *if we compare rounds that start from the same population count, does learning whether the controller advocated or did nothing improve our prediction of the next population count?*

The calculation proceeds through four layers:

$$\boxed{\text{microscopic population rule}} \implies \boxed{K_0, K_1} \implies \boxed{R_0, R_1} \implies \boxed{T(n)}.$$

The rest of this tutorial builds these objects one at a time.

2 State, timing, variables, and parameters

2.1 Population state

At the start of round k , let

$$N_k = n \in \{0, 1, \dots, N\}. \quad (3)$$

Thus n agents support A , and

$$N - n$$

agents support B .

It is often convenient to use the corresponding fraction

$$x_k = \frac{N_k}{N}, \quad x = \frac{n}{N} \in [0, 1]. \quad (4)$$

Interpretation:

$$x = 0 \Rightarrow \text{all } B, \quad x = \frac{1}{2} \Rightarrow \text{even split}, \quad x = 1 \Rightarrow \text{all } A.$$

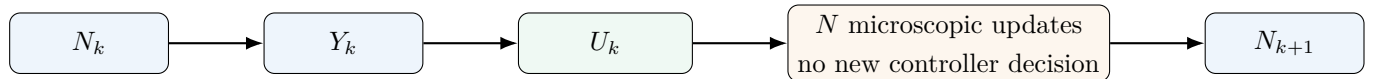
2.2 Symbol table

Symbol	Meaning
N	Population size. One population round contains exactly N microscopic focal-update positions.
q	Number of social inputs used by the unanimity q -voter response rule.
q_c	Number of agents sampled by the controller at the beginning of a round. This is a sensing resource, not the social-interaction order.
b	Exact number of microscopic update positions controlled during an advocacy round.
$c = b/N$	Fraction of microscopic positions controlled during an advocacy round.
N_k	Number of A supporters at the start of round k .
n	A particular value of N_k .
$x = n/N$	Fraction of A supporters corresponding to state n .
Y_k	Number of A supporters observed in the controller's sample.
y	A particular value of Y_k , with $0 \leq y \leq q_c$.
U_k	Controller action during round k : NoOp or Advocate.
θ	Intervention threshold in the controller's sigmoid policy.
β	Slope / decision stochasticity of the controller policy. Large β makes the policy more threshold-like.
K_0	One-microscopic-update transition kernel without controller advocacy.
K_1	One-microscopic-update transition kernel at a controlled position, where one social slot is replaced by a controller message supporting A .
R_0	Whole-round transition kernel under NoOp.
R_1	Whole-round transition kernel under Advocate.

Symbol	Meaning
$S(y n)$	Sensor law: probability that the controller sees y target supporters when the true population count is n .
a_n	State-conditioned advocacy probability $\Pr(U_k = \text{Advocate} N_k = n)$.
$M_n(m)$	Next-state distribution after marginalizing over the controller action.
$T(n)$	Exact state-local transfer entropy $I(U_k; N_{k+1} N_k = n)$.
$P_k(n)$	Distribution of the population count at the start of round k .
T_k	Round- k transfer entropy $I(U_k; N_{k+1} N_k)$.
$T^{(K)}$	Average transfer entropy over a finite K -round episode.
$f_0(x)$	Large- N ordinary microscopic drift in the count variable.
$f_1(x)$	Large- N controlled microscopic drift.
$\Delta f_q(x)$	Drift separation $f_1(x) - f_0(x)$.
$\nu_0(x)$	Ordinary microscopic variance coefficient.
J_k	One-round target current, $J_k = N_{k+1} - N_k$. Positive means net motion toward target A .
$J^{(K)}$	Finite-horizon episode current, $J^{(K)} = N_K - N_0$.
μ_J	Mean finite-horizon current, $\mathbb{E}[J^{(K)}]$.
V_J	Current variance, $\text{Var}(J^{(K)})$.
F_J^{disp}	Fano-like dispersion $V_J/ \mu_J $.
P_J^{Iri}	Inverse-Fano current precision $ \mu_J /V_J$. The superscript only labels the convention used in the social-imitation precision discussion.
SNR_J^2	Squared current signal-to-noise ratio μ_J^2/V_J .
$\mathcal{R}$	Whole-round kernel used to propagate a finite-horizon current experiment: R_0 , R_1 , or the feedback mixture R .
$P_0(n_0)$	Distribution of the initial target count used for finite-horizon current predictions.

2.3 Round timing

The timing is



If $U_k = \text{Advocate}$, a set

$$S_k \subset \{1, \dots, N\}, \quad |S_k| = b,$$

of microscopic *positions* is drawn uniformly before the round. The controller does not select b fixed agent identities. This keeps the dynamics exchangeable at the population level.

Common point of confusion

The controller acts *once per round*, not once per microscopic update. The N microscopic positions use the already-chosen round action. This is why the natural information channel is from U_k to N_{k+1} conditional on N_k .

3 Ordinary microscopic q -voter kernel K_0

A single microscopic update changes the number of A supporters by at most one:

$$n \longrightarrow n + 1, \quad n \longrightarrow n, \quad n \longrightarrow n - 1.$$

We derive all three probabilities.

3.1 Derivation of $K_0(n + 1 | n)$

For n to increase by one, a B agent must switch to A . Two independent selection requirements must both occur.

3.1.1 Requirement 1: the focal agent is a B supporter

There are $N - n$ such agents among N total agents, so

$$\Pr(\text{focal is } B | n) = \frac{N - n}{N}. \quad (5)$$

3.1.2 Requirement 2: all q sampled social inputs are A supporters

After the focal agent is removed, there are $N - 1$ possible neighbors. Because the focal agent is B , all n existing A supporters remain available among these $N - 1$ agents.

The total number of unordered groups of q neighbors is

$$\binom{N - 1}{q}.$$

The number of groups consisting entirely of A supporters is

$$\binom{n}{q}.$$

Therefore

$$\Pr(\text{all } q \text{ inputs are } A | \text{focal } B, n) = \frac{\binom{n}{q}}{\binom{N-1}{q}}. \quad (6)$$

3.1.3 Multiply the two requirements

Using Eqs. (5) and (6),

$$K_0(n+1 | n) = \underbrace{\frac{N-n}{N}}_{\text{choose focal } B} \underbrace{\frac{\binom{n}{q}}{\binom{N-1}{q}}}_{\text{all } q \text{ inputs are } A}. \quad (7)$$

Hence

Result

$$K_0(n+1 | n) = \frac{N-n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}}. \quad (8)$$

3.2 Derivation of $K_0(n-1 | n)$

Now an A supporter must switch to B .

The focal agent is A with probability

$$\frac{n}{N}.$$

If the focal agent is A , the other $N-1$ agents contain all $N-n$ B supporters. Unanimous B input occurs with probability

$$\frac{\binom{N-n}{q}}{\binom{N-1}{q}}.$$

Thus

Result

$$K_0(n-1 | n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}}. \quad (9)$$

3.3 The diagonal probability

The three possibilities exhaust all outcomes, so the row must sum to one:

$$K_0(n | n) = 1 - K_0(n+1 | n) - K_0(n-1 | n). \quad (10)$$

This completes the ordinary microscopic kernel.

4 Controlled microscopic kernel K_1

At a controlled position, the focal response rule remains a unanimity q -voter rule. The only change is that one of the q social-input slots is occupied by a controller message supporting A .

Thus the focal agent sees

1 controller input saying $A + (q - 1)$ ordinary inputs.

4.1 Derivation of $K_1(n + 1 | n)$

As before, the focal agent must be a B supporter, giving

$$\frac{N - n}{N}.$$

But one required A input has already been supplied by the controller. Only $q - 1$ ordinary A supporters are now required:

$$\frac{\binom{n}{q-1}}{\binom{N-1}{q-1}}.$$

Therefore

Result

$$K_1(n + 1 | n) = \frac{N - n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}}. \quad (11)$$

4.2 Why $K_1(n - 1 | n) = 0$

For an A focal agent to switch to B , all q inputs must be B . But one slot is fixed to the controller's A message. Unanimous B input is therefore impossible:

Result

$$K_1(n - 1 | n) = 0. \quad (12)$$

The diagonal term is

$$K_1(n | n) = 1 - K_1(n + 1 | n). \quad (13)$$

Intuition

Control works in two ways simultaneously: it makes $B \rightarrow A$ easier because only $q - 1$ ordinary A inputs are needed, and it blocks $A \rightarrow B$ at controlled positions because the controller's A message prevents unanimous B input.

5 Worked microscopic example: $N = 32$, $q = 2$, $n = 8$

This example makes the abstract kernels concrete.

We have

$$N = 32, \quad q = 2, \quad n = 8, \quad x = \frac{8}{32} = 0.25.$$

There are 24 B supporters.

5.1 Ordinary increase probability

From Eq. (8),

$$K_0(9 | 8) = \frac{24}{32} \frac{\binom{8}{2}}{\binom{31}{2}} \quad (14)$$

$$= \frac{3}{4} \frac{28}{465} \quad (15)$$

$$\approx 0.04516. \quad (16)$$

5.2 Ordinary decrease probability

From Eq. (9),

$$K_0(7 | 8) = \frac{8}{32} \frac{\binom{24}{2}}{\binom{31}{2}} \quad (17)$$

$$= \frac{1}{4} \frac{276}{465} \quad (18)$$

$$\approx 0.14839. \quad (19)$$

So the uncontrolled microscopic dynamics at this state favor movement toward B .

5.3 Controlled increase probability

For $q = 2$, a controlled position needs only $q - 1 = 1$ ordinary A supporter:

$$K_1(9 | 8) = \frac{24}{32} \frac{\binom{8}{1}}{\binom{31}{1}} \quad (20)$$

$$= \frac{3}{4} \frac{8}{31} \quad (21)$$

$$= \frac{6}{31} \quad (22)$$

$$\approx 0.19355. \quad (23)$$

Meanwhile,

$$K_1(7 | 8) = 0.$$

Logic of the step

The controlled input changes the one-step process from roughly

$$P(+1) = 0.045, \quad P(-1) = 0.148$$

to

$$P(+1) = 0.194, \quad P(-1) = 0.$$

The whole-round theory simply propagates these microscopic differences through N sequential update positions.

6 Controller sensing: the hypergeometric law

The controller does not observe n directly. At the start of the round it samples q_c agents without replacement and records

$$Y_k = y,$$

the number of sampled agents supporting A .

We derive

$$S(y | n) = \Pr(Y_k = y | N_k = n).$$

To obtain exactly y sampled A supporters:

- choose y agents from the n available A supporters: $\binom{n}{y}$ possibilities;
- choose the remaining $q_c - y$ agents from the $N - n$ B supporters: $\binom{N-n}{q_c-y}$ possibilities;
- divide by the number of all possible samples of size q_c : $\binom{N}{q_c}$.

Therefore

Result

$$S(y | n) = \Pr(Y_k = y | N_k = n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (24)$$

Common point of confusion

q and q_c are different parameters. q belongs to the population's social response rule. q_c belongs only to the controller's measurement process.

7 The soft round controller

Given sensor outcome y , the measured target fraction is

$$\frac{y}{q_c}.$$

The controller uses the sigmoid policy

$$\Pr(U_k = \text{Advocate} | Y_k = y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (25)$$

7.1 What do θ and β do?

θ is the intervention trigger. With $\theta = 1/2$, advocacy becomes more likely when the measured target fraction is below one half.

β controls how sharply the controller responds to the threshold:

$$\beta \rightarrow 0 \Rightarrow \sigma(\beta \cdot) \approx \frac{1}{2},$$

so the decision is nearly random, whereas large β makes the policy close to a deterministic threshold rule.

7.2 Average over sensor uncertainty

We need the advocacy probability conditioned on the *true* state n :

$$a_n \equiv \Pr(U_k = \text{Advocate} \mid N_k = n).$$

The controller does not see n , so we marginalize over all possible y :

$$a_n = \sum_y \Pr(Y_k = y \mid n) \Pr(\text{Advocate} \mid Y_k = y) \quad (26)$$

$$= \sum_y S(y \mid n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right]. \quad (27)$$

Thus

Result

$$a_n = \sum_y S(y \mid n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right]. \quad (28)$$

8 Exact simplification for $q_c = 2$ and $\theta = 1/2$

The source report highlights an exact simplification. We now derive it carefully.

When $q_c = 2$, the sensor outcome is only

$$y \in \{0, 1, 2\}.$$

Define

$$s_y = \Pr(\text{Advocate} \mid Y = y).$$

With $\theta = 1/2$,

$$s_0 = \sigma(\beta/2), \quad (29)$$

$$s_1 = \sigma(0) = \frac{1}{2}, \quad (30)$$

$$s_2 = \sigma(-\beta/2) = 1 - s_0. \quad (31)$$

Let

$$P_y = \Pr(Y = y \mid n).$$

Then

$$a_n = P_0 s_0 + P_1 \frac{1}{2} + P_2 (1 - s_0). \quad (32)$$

Because $P_0 + P_1 + P_2 = 1$,

$$a_n = \frac{1}{2} + \left(s_0 - \frac{1}{2}\right) (P_0 - P_2). \quad (33)$$

We simplify the two bracketed factors separately.

8.1 First factor: $s_0 - 1/2$

Use

$$\sigma(z) = \frac{1}{2} \left[1 + \tanh\left(\frac{z}{2}\right) \right]. \quad (34)$$

Therefore

$$s_0 - \frac{1}{2} = \sigma(\beta/2) - \frac{1}{2} \quad (35)$$

$$= \frac{1}{2} \tanh\left(\frac{\beta}{4}\right). \quad (36)$$

8.2 Second factor: $P_0 - P_2$

The hypergeometric probabilities are

$$P_0 = \frac{\binom{N-n}{2}}{\binom{N}{2}}, \quad P_2 = \frac{\binom{n}{2}}{\binom{N}{2}}.$$

Hence

$$P_0 - P_2 = \frac{\binom{N-n}{2} - \binom{n}{2}}{\binom{N}{2}} \quad (37)$$

$$= \frac{(N-n)(N-n-1) - n(n-1)}{N(N-1)} \quad (38)$$

$$= \frac{N^2 - 2Nn + n^2 - N + n - (n^2 - n)}{N(N-1)} \quad (39)$$

$$= \frac{N^2 - 2Nn - N + 2n}{N(N-1)} \quad (40)$$

$$= \frac{(N-1)(N-2n)}{N(N-1)} \quad (41)$$

$$= 1 - \frac{2n}{N} \quad (42)$$

$$= 1 - 2x. \quad (43)$$

Substitute Eqs. (36) and (43) into Eq. (33):

Result

$$a(x) = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{\beta}{4}\right) (1 - 2x). \quad (44)$$

For $\beta = 4$,

$$\tanh(1) \approx 0.761594,$$

so

$$a(x) = \frac{1}{2} + \frac{1}{2} (0.761594) (1 - 2x) \quad (45)$$

$$= 0.880797 - 0.761594x. \quad (46)$$

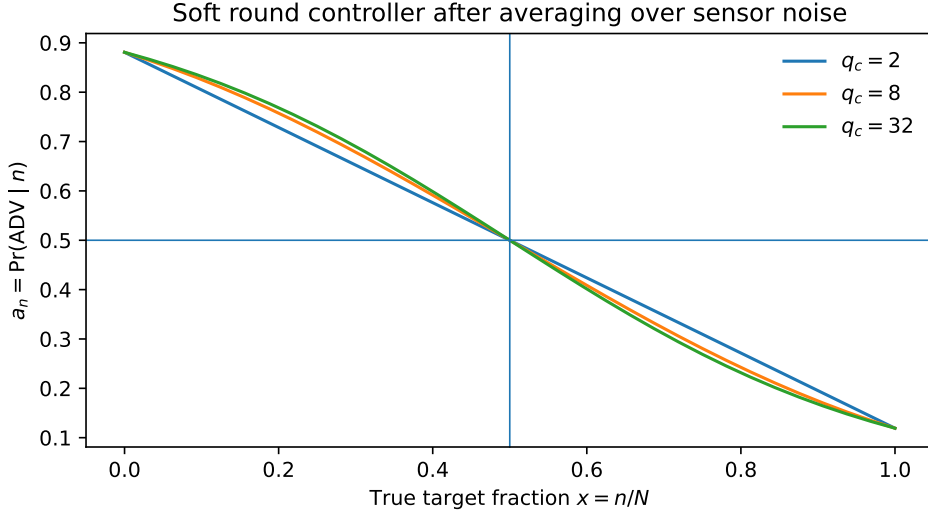


Figure 1: State-conditioned advocacy probability after averaging over the finite sensor for $N = 32$, $\theta = 1/2$, $\beta = 4$. This reproduces the source-note calculation. Changing q_c changes sensing fidelity but not the q -voter population rule.

9 Whole-round kernels: from microscopic updates to N_{k+1}

The transfer entropy in Eq. (1) compares states from one round boundary to the next. Therefore K_0 and K_1 , which describe only one microscopic update, must be propagated through all N positions.

9.1 NoOp round: $R_0 = K_0^N$

Under NoOp, all N positions use K_0 . In matrix notation,

Result

$$R_0 = K_0^N. \quad (47)$$

Why does matrix multiplication do the correct averaging? For two updates,

$$[K_0^2]_{n,m} = \sum_j K_0(j | n) K_0(m | j). \quad (48)$$

The sum runs over every possible intermediate population count j . Repeated multiplication automatically sums over every length- N microscopic trajectory.

9.2 Why R_1 is not simply $K_1^b K_0^{N-b}$

In an advocacy round, exactly b positions are controlled, but their positions are uniformly preallocated. For example, with $N = 6$ and $b = 2$, possible schedules include

$$K_1 K_0 K_0 K_1 K_0 K_0, \quad K_0 K_1 K_0 K_0 K_1 K_0, \quad \dots$$

There are $\binom{N}{b}$ such schedules. A fixed product $K_1^b K_0^{N-b}$ would represent only one ordering.

9.3 Dynamic-programming state $F_{r,j}$

Define $F_{r,j}$ as the transition matrix after

- r positions have already been processed;
- exactly j of those r positions were controlled.

Initially,

$$F_{0,0} = I. \quad (49)$$

At stage (r, j) there are

$$N - r$$

positions remaining. We still need

$$b - j$$

controlled positions. Therefore

$$\Pr(\text{next position controlled} \mid r, j) = \frac{b - j}{N - r}. \quad (50)$$

How many ordinary positions remain? The total number of ordinary positions is $N - b$. We have already used $r - j$ ordinary positions, so the remaining number is

$$(N - b) - (r - j) = N - b - r + j. \quad (51)$$

Hence

$$\Pr(\text{next position ordinary} \mid r, j) = \frac{N - b - r + j}{N - r}. \quad (52)$$

The two probabilities sum to one exactly.

9.4 The recursion

If the next position is controlled, append K_1 :

$$F_{r+1,j+1} += \frac{b - j}{N - r} F_{r,j} K_1. \quad (53)$$

If it is ordinary, append K_0 :

$$F_{r+1,j} += \frac{N - b - r + j}{N - r} F_{r,j} K_0. \quad (54)$$

After all N positions, exactly b controlled positions must have been used, so

Result

$$R_1 = F_{N,b}. \quad (55)$$

Logic of the step

The recursion is simply a compact way to average over every uniformly chosen schedule of b controlled positions. It gives exactly the same result as explicitly summing the $\binom{N}{b}$ schedule-specific matrix products, but without enumerating them.

10 Exact finite- N transfer entropy

We now have the two next-round laws at fixed current state n :

$$R_0(m | n) = \Pr(N_{k+1} = m | N_k = n, U_k = \text{NoOp}), \quad (56)$$

$$R_1(m | n) = \Pr(N_{k+1} = m | N_k = n, U_k = \text{Advocate}). \quad (57)$$

The action probabilities are

$$\Pr(\text{Advocate} | n) = a_n, \quad \Pr(\text{NoOp} | n) = 1 - a_n.$$

10.1 Marginalize over the hidden controller action

Suppose we know the current state n but not the controller action. The next-state law follows from the law of total probability:

$$M_n(m) \equiv \Pr(N_{k+1} = m | N_k = n) \quad (58)$$

$$= \sum_u \Pr(u | n) \Pr(m | n, u) \quad (59)$$

$$= (1 - a_n)R_0(m | n) + a_n R_1(m | n). \quad (60)$$

Thus

Result

$$M_n(m) = (1 - a_n)R_0(m | n) + a_n R_1(m | n). \quad (61)$$

10.2 Start from conditional mutual information

At fixed n ,

$$T(n) \equiv I(U_k; N_{k+1} | N_k = n). \quad (62)$$

For generic discrete variables U and M ,

$$I(U; M) = \sum_{u,m} p(u, m) \log_2 \frac{p(u, m)}{p(u)p(m)}. \quad (63)$$

Conditioning on n and factorizing

$$p(u, m | n) = p(u | n)p(m | n, u),$$

we get

$$T(n) = \sum_{u, m} p(u | n)p(m | n, u) \log_2 \frac{p(m | n, u)}{p(m | n)}. \quad (64)$$

Now insert the two actions.

For $u = \text{NoOp}$,

$$p(u | n) = 1 - a_n, \quad p(m | n, u) = R_0(m | n).$$

For $u = \text{Advocate}$,

$$p(u | n) = a_n, \quad p(m | n, u) = R_1(m | n).$$

And by [Eq. \(61\)](#),

$$p(m | n) = M_n(m).$$

Therefore

Result
$T(n) = (1 - a_n) \sum_m R_0(m n) \log_2 \frac{R_0(m n)}{M_n(m)} + a_n \sum_m R_1(m n) \log_2 \frac{R_1(m n)}{M_n(m)} \quad (65)$

This is an exact finite-population result. Once $(N, q, q_c, b, \beta, \theta)$ are fixed, no empirical mutual-information estimator appears anywhere in the calculation.

11 Why the exact TE is a weighted Jensen–Shannon divergence

The Kullback–Leibler divergence is

$$D_{\text{KL}}(P \| Q) = \sum_m P(m) \log_2 \frac{P(m)}{Q(m)}. \quad (66)$$

Thus [Eq. \(65\)](#) becomes

$$T(n) = (1 - a_n) D_{\text{KL}}(R_0 \| M_n) + a_n D_{\text{KL}}(R_1 \| M_n), \quad (67)$$

with

$$M_n = (1 - a_n) R_0 + a_n R_1.$$

This is exactly the weighted Jensen–Shannon divergence:

Result

$$T(n) = \text{JS}_{a_n}(R_0(\cdot | n), R_1(\cdot | n)). \quad (68)$$

Intuition

The Jensen–Shannon form gives the cleanest interpretation of the theory: at a fixed current population state, transfer entropy is the statistical distinguishability of the complete next-round distributions produced by NoOp and Advocate.

11.1 Two ingredients required for positive TE

The Jensen–Shannon form shows immediately that two separate ingredients are necessary.

1. Action variation at fixed state. We need

$$0 < a_n < 1.$$

If $a_n = 0$ or $a_n = 1$, the action is deterministic at that state and contains no uncertainty to transmit.

2. Dynamical separation. We need

$$R_0(\cdot | n) \neq R_1(\cdot | n).$$

If the two actions generate the same next-state law, knowing the action does not help us predict the future.

Thus

$$\text{positive TE} = \text{action variation} + \text{action-dependent dynamics}.$$

12 From local TE to round and episode quantities

Let

$$P_k(n) = \Pr(N_k = n)$$

be the population-state distribution at the start of round k . Then

Result

$$T_k = I(U_k; N_{k+1} | N_k) = \sum_n P_k(n) T(n). \quad (69)$$

The population distribution itself propagates through the action-marginalized kernel:

$$P_{k+1}(m) = \sum_n P_k(n) M_n(m) \quad (70)$$

$$= \sum_n P_k(n) [(1 - a_n) R_0(m | n) + a_n R_1(m | n)]. \quad (71)$$

Starting from P_0 , Eq. (71) determines $P_1, P_2, \dots$, and Eq. (69) then determines the round-by-round information transfer.

For a K -round episode, the finite-horizon average is

Result

$$T^{(K)} = \frac{1}{K} \sum_{k=0}^{K-1} T_k. \quad (72)$$

Common point of confusion

The pure unanimity voter has absorbing consensus states. Long-time stationary TE can therefore vanish after the process reaches consensus. For comparison with finite multi-agent LLM episodes, the state-local $T(n)$, round-local T_k , and finite-horizon $T^{(K)}$ are usually more informative than an asymptotic stationary rate.

13 How the exact theory curves are generated

For every state $n = 0, \dots, N$ and every chosen parameter tuple $(N, q, q_c, b, \beta, \theta)$:

1. Construct the microscopic matrices K_0 and K_1 from Eqs. (8), (9), (11) and (12).
2. Compute $R_0 = K_0^N$.
3. Compute $R_1 = F_{N,b}$ using Eqs. (53) and (54).
4. Compute the controller probability a_n from Eq. (28).
5. Form M_n using Eq. (61).
6. Evaluate $T(n)$ using Eq. (65).
7. Plot $T(n)$ against $x = n/N$.

The following curves reproduce the source-note construction for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$.

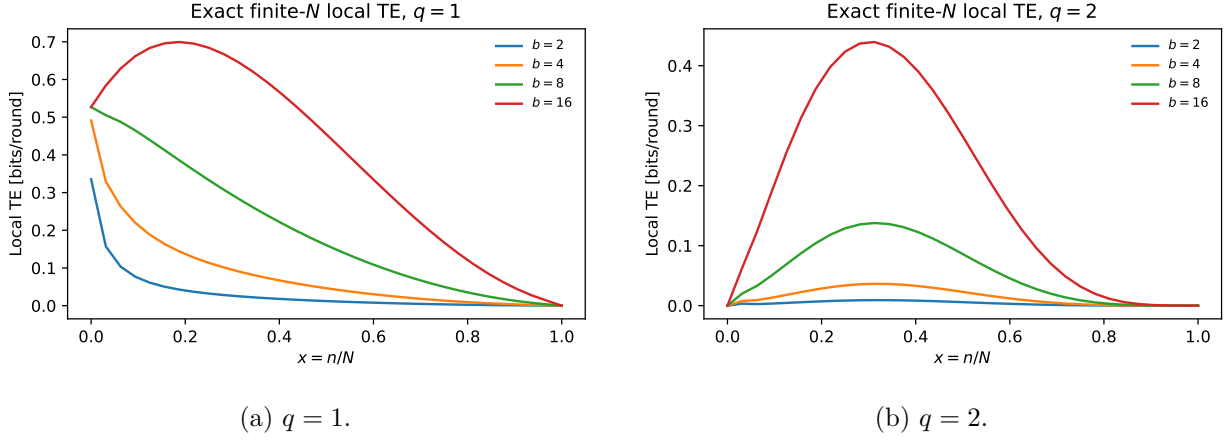


Figure 2: Exact finite- N state-local transfer entropy for several round budgets b . These are theory curves obtained from $K_0, K_1 \rightarrow R_0, R_1 \rightarrow T(n)$, not Monte Carlo estimates.

14 Large- N approximation: why we introduce it

The exact finite- N calculation tells us the correct answer, but its matrix recursion hides the scaling structure. To understand how the information channel depends on q , b/N , and intrinsic stochastic noise, we approximate

$$x = \frac{n}{N}$$

as continuous and derive a local drift/noise description.

15 Large- N ordinary and controlled drifts

15.1 Ordinary q -voter transition probabilities

For large N ,

$$\frac{\binom{n}{q}}{\binom{N-1}{q}} \approx x^q, \quad \frac{\binom{N-n}{q}}{\binom{N-1}{q}} \approx (1-x)^q.$$

Therefore the approximate probability that a single ordinary update changes the count by $+1$ is

$$p_+(x) = (1-x)x^q, \tag{73}$$

and the probability of a -1 change is

$$p_-(x) = x(1-x)^q. \tag{74}$$

Let the microscopic count increment be

$$Z \in \{-1, 0, +1\}.$$

Then

$$\mathbb{E}[Z] = (+1)p_+ + (-1)p_- + 0 \cdot p_0 \quad (75)$$

$$= p_+ - p_-. \quad (76)$$

Thus

Result

$$f_0(x) = (1-x)x^q - x(1-x)^q. \quad (77)$$

15.2 Controlled drift

At a controlled position, a B focal agent needs only $q - 1$ ordinary A supporters because the controller supplies one A input. Hence

$$p_+^{(1)}(x) = (1-x)x^{q-1}.$$

A controlled $A \rightarrow B$ switch is impossible, so

$$p_-^{(1)}(x) = 0.$$

Therefore

Result

$$f_1(x) = (1-x)x^{q-1}. \quad (78)$$

15.3 Control-induced drift separation

Define

$$\Delta f_q(x) = f_1(x) - f_0(x).$$

Substitute Eqs. (77) and (78):

$$\Delta f_q(x) = (1-x)x^{q-1} - [(1-x)x^q - x(1-x)^q] \quad (79)$$

$$= (1-x)x^{q-1} - (1-x)x^q + x(1-x)^q \quad (80)$$

$$= (1-x)x^{q-1} \underbrace{(1-x)}_{1-x} + x(1-x)^q \quad (81)$$

$$= x^{q-1}(1-x)^2 + x(1-x)^q. \quad (82)$$

Thus

Result

$$\Delta f_q(x) = x^{q-1}(1-x)^2 + x(1-x)^q. \quad (83)$$

16 Ordinary microscopic noise coefficient

The variance of Z is

$$\text{Var}(Z) = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2. \quad (84)$$

Because $Z^2 = 1$ for both $+1$ and -1 jumps,

$$\mathbb{E}[Z^2] = p_+ + p_-.$$

Since $\mathbb{E}[Z] = f_0(x)$,

$$\nu_0(x) \equiv \text{Var}(Z) \quad (85)$$

$$= p_+(x) + p_-(x) - f_0(x)^2 \quad (86)$$

$$= (1-x)x^q + x(1-x)^q - f_0(x)^2. \quad (87)$$

Hence

Result

$$\nu_0(x) = (1-x)x^q + x(1-x)^q - f_0(x)^2. \quad (88)$$

17 From microscopic drift to whole-round mean separation

A one-unit change in count n corresponds to a change

$$\Delta x = \frac{1}{N}$$

in the population fraction.

An advocacy round contains exactly

$$b = cN$$

controlled positions. Relative to an ordinary position, one controlled position changes the expected count increment by approximately $\Delta f_q(x)$. In the fraction variable, this is

$$\frac{\Delta f_q(x)}{N}.$$

Across cN controlled positions,

$$\Delta\mu(x) \approx cN \frac{\Delta f_q(x)}{N} \quad (89)$$

$$= c\Delta f_q(x). \quad (90)$$

Therefore

Result
$\Delta\mu(x) \approx c\Delta f_q(x). \quad (91)$

This is the approximate separation between the mean next-round fractions under Advocate and NoOp.

18 Whole-round ordinary fluctuations

For one ordinary update,

$$\text{Var}(\Delta x) = \frac{\nu_0(x)}{N^2}.$$

Over N microscopic positions, the local diffusion approximation gives

$$s^2(x) \approx N \frac{\nu_0(x)}{N^2} \quad (92)$$

$$= \frac{\nu_0(x)}{N}. \quad (93)$$

Hence

Result
$s^2(x) \approx \frac{\nu_0(x)}{N}. \quad (94)$

This is the key finite-size scaling: fluctuations of the *fraction* shrink like $N^{-1/2}$.

19 Second-order Jensen–Shannon expansion

Now approximate the two round kernels locally by distributions with the same variance s^2 but nearby means:

$$R_0 \approx \mathcal{N}(\mu_0, s^2), \quad R_1 \approx \mathcal{N}(\mu_1, s^2), \quad \delta \equiv \mu_1 - \mu_0.$$

The weighted mixture is

$$M = (1 - a)R_0 + aR_1.$$

For small δ , the KL divergence between nearby parametric distributions has the quadratic expansion

$$D_{\text{KL}}(p_\mu \| p_{\mu+\epsilon}) = \frac{1}{2} I_F(\mu) \epsilon^2 + O(\epsilon^3), \quad (95)$$

where I_F is the Fisher information.

The mixture mean is approximately

$$\mu_M = (1 - a)\mu_0 + a\mu_1 = \mu_0 + a\delta.$$

Thus the displacement from R_0 to M is approximately $a\delta$, while the displacement from R_1 to M is approximately $(1 - a)\delta$. Therefore, in natural-log units,

$$D_{\text{KL}}(R_0 \| M) \approx \frac{1}{2} I_F a^2 \delta^2, \quad (96)$$

$$D_{\text{KL}}(R_1 \| M) \approx \frac{1}{2} I_F (1 - a)^2 \delta^2. \quad (97)$$

Insert these into the weighted Jensen–Shannon divergence:

$$\text{JS}_a(R_0, R_1) \approx (1 - a) \frac{1}{2} I_F a^2 \delta^2 + a \frac{1}{2} I_F (1 - a)^2 \delta^2 \quad (98)$$

$$= \frac{1}{2} I_F \delta^2 \left[(1 - a)a^2 + a(1 - a)^2 \right] \quad (99)$$

$$= \frac{1}{2} I_F \delta^2 [a(1 - a)(a + 1 - a)] \quad (100)$$

$$= \frac{1}{2} a(1 - a) I_F \delta^2. \quad (101)$$

For a Gaussian mean parameter,

$$I_F = \frac{1}{s^2}.$$

So in nats,

$$\text{JS}_a \approx \frac{a(1 - a)}{2} \frac{\delta^2}{s^2}.$$

The exact TE in the report uses base-2 logarithms, so divide by $\ln 2$:

Result

$$T \approx \frac{a(1 - a)}{2 \ln 2} \frac{\delta^2}{s^2}. \quad (102)$$

20 Deriving the reported mean-field transfer entropy

Use

$$\delta \approx c \Delta f_q(x)$$

from Eq. (91) and

$$s^2 \approx \frac{\nu_0(x)}{N}$$

from Eq. (94). Then

$$\frac{\delta^2}{s^2} \approx \frac{c^2 [\Delta f_q(x)]^2}{\nu_0(x)/N} \quad (103)$$

$$= Nc^2 \frac{[\Delta f_q(x)]^2}{\nu_0(x)}. \quad (104)$$

Substitute into Eq. (102):

Result

$$T_{\text{MF}}(x) \approx \frac{a(x)[1-a(x)]}{2 \ln 2} Nc^2 \frac{[\Delta f_q(x)]^2}{\nu_0(x)}. \quad (105)$$

This is the closed large- N , weak-control approximation reported in the source note.

21 Reading the mean-field formula factor by factor

Equation Eq. (105) becomes much easier to understand if we separate its components:

$$T_{\text{MF}} \approx \underbrace{a(1-a)}_{\text{action variation}} \underbrace{\frac{1}{2 \ln 2}}_{\text{quadratic JS in bits}} \underbrace{Nc^2}_{\text{actuation strength}} \underbrace{\frac{\Delta f_q^2}{\nu_0}}_{\text{dynamical response to noise}}.$$

21.1 Action variation: $a(1-a)$

This factor is maximal at $a = 1/2$ and vanishes at $a = 0$ or $a = 1$:

$$a(1-a) \leq \frac{1}{4}.$$

A completely deterministic action at fixed state leaves no action uncertainty to transmit.

21.2 Quadratic control-budget scaling: c^2

The mean displacement is linear in control fraction:

$$\Delta\mu \propto c.$$

But local statistical distinguishability is quadratic in a small displacement:

$$T \propto (\Delta\mu)^2.$$

Hence

$$T \propto c^2 \quad \text{for weak control.}$$

21.3 Population size: N

For the fraction x , intrinsic round-to-round noise scales like

$$s \sim N^{-1/2}.$$

At fixed c , a control-induced displacement therefore becomes easier to distinguish in larger populations.

21.4 Response-to-noise ratio

The ratio

$$\frac{[\Delta f_q(x)]^2}{\nu_0(x)} \quad (106)$$

compares the squared change in deterministic drift caused by control with the ordinary stochastic variance. It acts like a local dynamical signal-to-noise ratio.

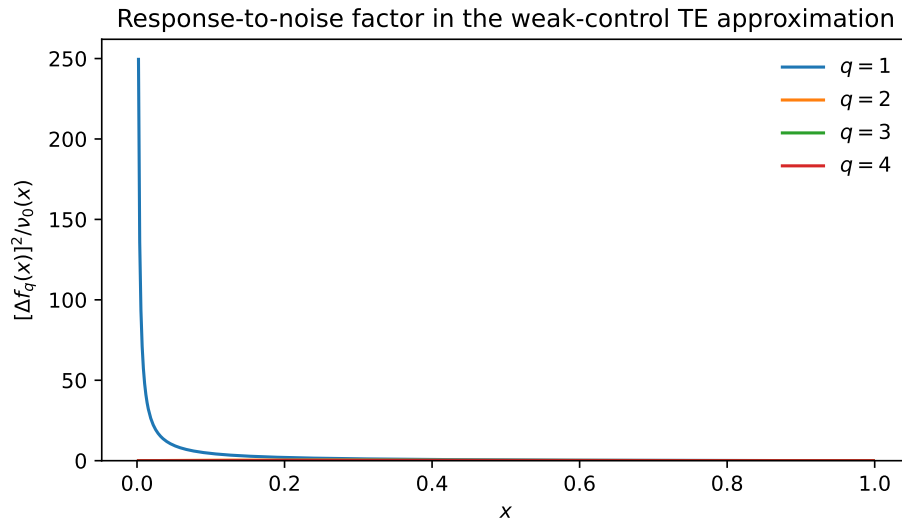


Figure 3: Tutorial-only diagnostic derived directly from the source formulas: the response-to-noise factor $[\Delta f_q(x)]^2/\nu_0(x)$ for several q . This plot was not part of the original five-page note; it is included to make the structure of Eq. (105) visually transparent.

22 Why the weak-control scaling can be written as b^2/N

Because

$$c = \frac{b}{N},$$

we have

$$Nc^2 = N \left(\frac{b}{N} \right)^2 \quad (107)$$

$$= \frac{b^2}{N}. \quad (108)$$

Therefore the characteristic scaling can also be written

$$T_{\text{MF}} \propto a(1-a) \frac{b^2}{N} \quad (109)$$

for fixed local response-to-noise factor.

Common point of confusion

Be careful about what is held fixed. If the absolute budget b is fixed while N grows, the signal weakens like $1/N$. If instead the fraction $c = b/N$ is fixed, then $b = cN$ and the same factor is Nc^2 , which grows with N within the local weak-control approximation.

23 The analytically stronger special case $q = 1$

For $q = 1$, the ordinary voter model has a particularly simple mean behavior.

From Eq. (77),

$$f_0(x) = (1-x)x - x(1-x) \quad (110)$$

$$= 0. \quad (111)$$

Thus

Result

$$f_0(x) = 0 \quad (q = 1). \quad (112)$$

Ordinary copying has zero mean drift.

At a controlled position, the controller occupies the only social slot. Every selected B focal agent therefore becomes A . The probability of selecting B is $1 - x$, so

$$f_1(x) = 1 - x. \quad (113)$$

23.1 Exact expectation after b controlled positions

It is useful to make this derivation slightly more rigorous because the b controlled positions are interspersed with ordinary voter updates.

Let

$$B = N - n$$

be the number of non-target supporters. Under an ordinary $q = 1$ voter update, the target count is a martingale, so the conditional expected B count is unchanged:

$$\mathbb{E}[B' \mid B, \text{ordinary}] = B. \quad (114)$$

At a controlled position the controller occupies the only social slot. A B focal agent is converted to A whenever it is selected. Since the probability of selecting a B focal agent is B/N ,

$$\mathbb{E}[B' \mid B, \text{controlled}] = B - \frac{B}{N} \cdot 1 \quad (115)$$

$$= B \left(1 - \frac{1}{N}\right). \quad (116)$$

Logic of the step

The key point is that both conditional expectations are *linear* in B . Therefore the tower property lets us propagate expectations through any ordering of ordinary and controlled updates. Ordinary positions multiply the expected B count by 1; each controlled position multiplies it by $1 - 1/N$. Only the number of controlled positions matters for the mean.

After exactly b controlled positions,

$$\mathbb{E}[B_{k+1} \mid B_k = B, \text{Advocate}] = B \left(1 - \frac{1}{N}\right)^b. \quad (117)$$

Dividing by N gives the expected surviving non-target fraction,

$$\mathbb{E}[1 - x_{k+1} \mid x_k = x, \text{Advocate}] = (1 - x) \left(1 - \frac{1}{N}\right)^b. \quad (118)$$

Therefore

Result

$$\mathbb{E}[x_{k+1} \mid x_k = x, \text{Advocate}] = 1 - (1 - x) \left(1 - \frac{1}{N}\right)^b. \quad (119)$$

Under NoOp, the martingale property gives

$$\mathbb{E}[x_{k+1} \mid x_k = x, \text{NoOp}] = x. \quad (120)$$

Thus the exact mean separation is

$$\Delta\mu_{q=1}(x) = 1 - (1 - x) \left(1 - \frac{1}{N}\right)^b - x = (1 - x) \left[1 - \left(1 - \frac{1}{N}\right)^b\right]. \quad (121)$$

Hence

Result

$$\Delta\mu_{q=1}(x) = (1 - x) \left[1 - \left(1 - \frac{1}{N}\right)^b\right]. \quad (122)$$

In count rather than fraction units, the exact one-round mean current is therefore

Result

$$\mu_{J,1}^{(q=1)}(n) = (N - n) \left[1 - \left(1 - \frac{1}{N} \right)^b \right], \quad \mu_{J,0}^{(q=1)}(n) = 0. \quad (123)$$

The two formulas contain the same information because $J = N(x_{k+1} - x_k)$.

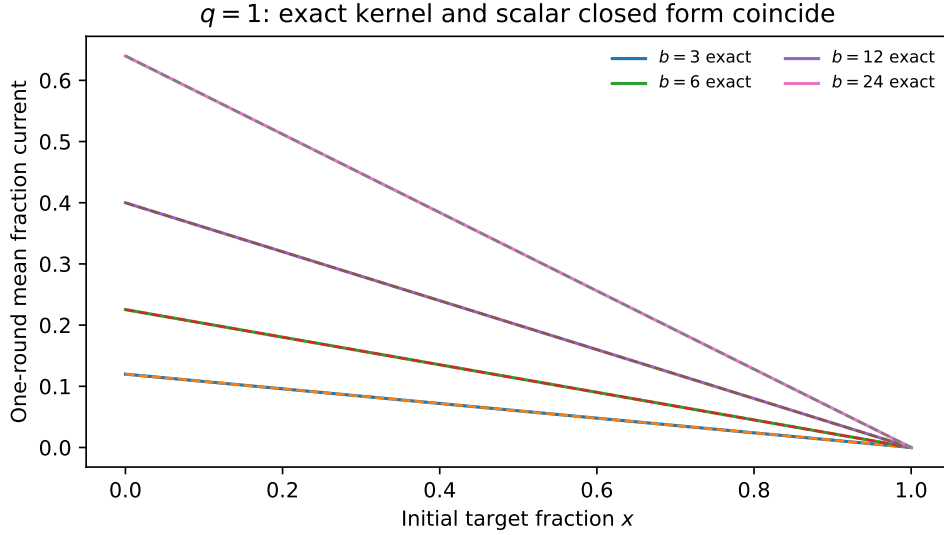


Figure 4: Exact one-round target-fraction current for $N = 24$, $q = 1$, and several control budgets. The solid curves are obtained from the exact whole-round kernel R_1 ; the dashed curves evaluate the scalar closed form in Eq. (122). Their overlap is an internal theory check, not a fit.

23.2 Large- N limit with $b = cN$

Use

$$\left(1 - \frac{1}{N} \right)^{cN} = \left[\left(1 - \frac{1}{N} \right)^N \right]^c \rightarrow e^{-c}.$$

Therefore

Result

$$\Delta\mu_{q=1}(x) \rightarrow (1 - x)(1 - e^{-c}). \quad (124)$$

This retains the full nonlinear dependence on c and is stronger than a first-order weak-control approximation.

24 The qualitative boundary distinction

Now set $x = 0$, meaning no ordinary A supporter exists.

24.1 $q = 1$: the controller can seed A from zero

At a controlled position the controller itself provides the only required social input. Therefore a selected B agent can switch to A even when $n = 0$. Under NoOp, however, the all- B state cannot leave itself. Therefore

$$R_0(\cdot | 0) \neq R_1(\cdot | 0),$$

so, provided $0 < a_0 < 1$,

$$\boxed{T(0) > 0 \quad (q = 1, b > 0).} \quad (125)$$

24.2 $q \geq 2$: one controller input is not enough

For $q \geq 2$, a controlled $B \rightarrow A$ switch still requires $q - 1 \geq 1$ ordinary A supporters. At $n = 0$ there are none:

$$\binom{0}{q-1} = 0.$$

Therefore

$$K_1(1 | 0) = 0.$$

Both NoOp and Advocate remain at the all- B boundary, so

$$R_0(\cdot | 0) = R_1(\cdot | 0),$$

and hence

$$\boxed{T(0) = 0 \quad (q \geq 2).} \quad (126)$$

The source note summarizes this distinction as

Result
$\boxed{q = 1 : T(0) > 0 \text{ for } b > 0, \quad q \geq 2 : T(0) = 0.} \quad (127)$

This explains the very different boundary behavior in Fig. 2.

25 Control-budget dependence and saturation

For each budget b , define the largest state-local information channel

$$T_{\max}(b) = \max_n T(n; b). \quad (128)$$

The exact finite- N curves are obtained by recomputing R_1 and $T(n)$ for each b .

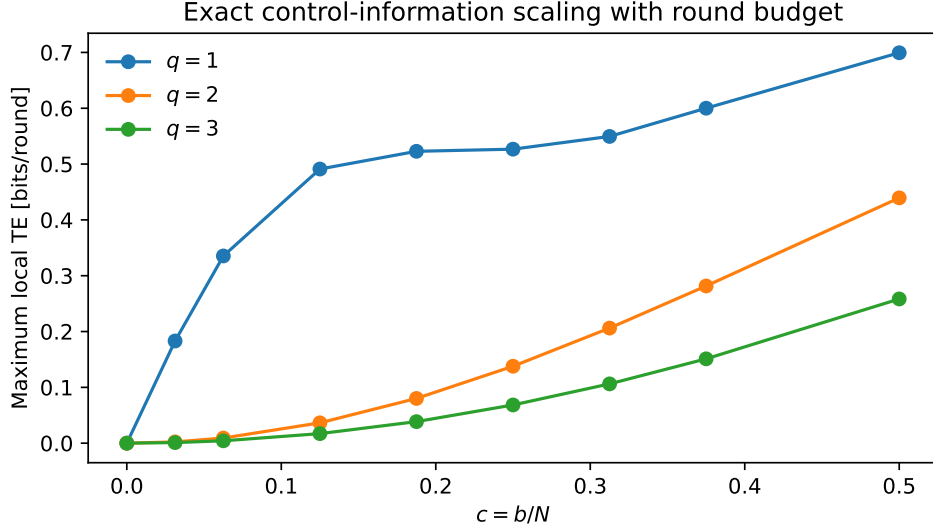


Figure 5: Maximum exact state-local transfer entropy over population states versus $c = b/N$ for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$. For $q \geq 2$, the low-budget behavior is consistent with the c^2 weak-control scaling. The $q = 1$ curve departs earlier because advocacy opens otherwise inaccessible boundary transitions.

Why does the exact curve eventually stop being quadratic? The derivation of Eq. (105) assumed that the two kernels are close enough for a second-order expansion. When b becomes large, R_0 and R_1 are no longer nearby distributions. The small-separation Taylor approximation ceases to be accurate, and information growth begins to saturate because the two actions are already highly distinguishable.

Thus the qualitative sequence is

$$\text{weak control} \longrightarrow c^2 \text{ regime} \longrightarrow \text{nonlinear separation} \longrightarrow \text{saturation.}$$

26 A second observable family: finite-horizon population current

The transfer-entropy calculation tells us how distinguishable the population dynamics are under different controller actions. We now use the *same* exact kernels to ask a more directly behavioral question: how far does the population move toward the controller target during a complete episode?

Let an episode contain K population rounds. Define

$$J^{(K)} = N_K - N_0. \quad (129)$$

We will usually call this simply the *target current*. It is positive if the target gains supporters, negative if it loses supporters, and zero if the final and initial target counts coincide.

Intuition

Suppose an episode starts with 6 of 24 agents supporting the target and ends with 18. Then $J = 18 - 6 = 12$. The current is not the final accuracy or final target share; it is the *change* in target occupancy. That distinction matters when different tasks or repetitions start from different initial states.

26.1 Microscopic current increments and the telescoping identity

At a single microscopic update t , define

$$j_t = \begin{cases} +1, & \text{if the focal agent switches from } B \text{ to target } A, \\ -1, & \text{if the focal agent switches from target } A \text{ to } B, \\ 0, & \text{if the target count does not change.} \end{cases} \quad (130)$$

If N_t denotes the target count immediately before that microscopic update, then simply

$$j_t = N_{t+1} - N_t. \quad (131)$$

Summing over all microscopic updates in an episode gives

$$J = \sum_t (N_{t+1} - N_t) \quad (132)$$

$$= (N_1 - N_0) + (N_2 - N_1) + \cdots + (N_T - N_{T-1}) \quad (133)$$

$$= N_T - N_0. \quad (134)$$

All intermediate counts cancel. This is called a *telescoping sum*.

Logic of the step

The microscopic definition and the endpoint definition are exactly equivalent. In simulation post-processing this gives a useful consistency check: summing all signed target-entry and target-exit events must reproduce the final target count minus the initial target count.

26.2 One-round current moments directly from a round kernel

Suppose a round starts from $N_k = n$. Under action $u \in \{0, 1\}$, the whole-round kernel $R_u(m | n)$ gives the probability of ending at $N_{k+1} = m$. The one-round current is

$$J_k = m - n.$$

Therefore the conditional mean current is just the expectation of $m - n$ under that row of the kernel:

$$\mu_{J,u}(n) = \mathbb{E}[J_k | N_k = n, U_k = u] = \sum_{m=0}^N (m - n) R_u(m | n). \quad (135)$$

The second moment is

$$M_{2,J,u}(n) = \mathbb{E}[J_k^2 \mid n, u] = \sum_{m=0}^N (m - n)^2 R_u(m \mid n). \quad (136)$$

Hence

$$V_{J,u}(n) = \text{Var}(J_k \mid n, u) = M_{2,J,u}(n) - \mu_{J,u}(n)^2. \quad (137)$$

Common point of confusion

These are exact finite- N expressions, but except in special cases such as the $q = 1$ mean they are not simple scalar closed forms. “Exact” here means that once R_u is built, the sum is evaluated without Monte Carlo approximation.

The action-induced current response is

$$\Delta\mu_J(n) = \mu_{J,1}(n) - \mu_{J,0}(n). \quad (138)$$

For $q = 1$, Eq. (138) reduces exactly to the closed form already derived in Eq. (123).

27 From one round to a complete episode

The LLM simulations we ultimately want to compare with the theory run for a finite number of population rounds, for example $K = 10$. The appropriate theoretical comparison is therefore also finite horizon.

27.1 Choose the correct whole-round propagator

Define $\mathcal{R}$ as the kernel actually governing one population round in the theoretical reference.

For an always-NoOp experiment,

$$\mathcal{R} = R_0. \quad (139)$$

For an always-Advocate experiment,

$$\mathcal{R} = R_1. \quad (140)$$

For a stochastic feedback controller, the action is marginalized conditional on the current state:

$$R(m \mid n) = [1 - a_n]R_0(m \mid n) + a_n R_1(m \mid n), \quad (141)$$

and then $\mathcal{R} = R$.

Intuition

R_0 and R_1 answer “what happens if I know the action?” The closed-loop kernel R answers “what happens if the controller follows its stochastic policy and I average over which action it happens to choose?” Both are useful, but for different questions.

27.2 Why powers of the kernel appear

If the initial target count is exactly n_0 , then after one round

$$\Pr(N_1 = m \mid N_0 = n_0) = \mathcal{R}(m \mid n_0).$$

After two rounds we must sum over every possible intermediate state r :

$$\Pr(N_2 = m \mid N_0 = n_0) = \sum_r \Pr(N_2 = m \mid N_1 = r) \Pr(N_1 = r \mid N_0 = n_0) \quad (142)$$

$$= \sum_r \mathcal{R}(m \mid r) \mathcal{R}(r \mid n_0). \quad (143)$$

In matrix notation this is the (n_0, m) element of $\mathcal{R}^2$. Repeating the argument gives

Result

$$\Pr(N_K = m \mid N_0 = n_0) = [\mathcal{R}^K]_{n_0 m}. \quad (144)$$

If the experiment has an initial distribution $P_0(n_0)$ rather than one fixed count, then

$$\Pr(N_K = m) = \sum_{n_0} P_0(n_0) [\mathcal{R}^K]_{n_0 m}. \quad (145)$$

28 The exact finite-horizon current distribution

Because

$$J^{(K)} = N_K - N_0,$$

the distribution of the current is already contained in Eq. (144).

For a fixed initial count n_0 , the event $J = j$ is the same event as $N_K = n_0 + j$. Therefore

Result

$$\Pr(J^{(K)} = j \mid N_0 = n_0) = [\mathcal{R}^K]_{n_0, n_0 + j}, \quad (146)$$

whenever $0 \leq n_0 + j \leq N$.

For a random initial count,

$$\Pr(J^{(K)} = j) = \sum_{n_0} P_0(n_0) [\mathcal{R}^K]_{n_0, n_0 + j}. \quad (147)$$

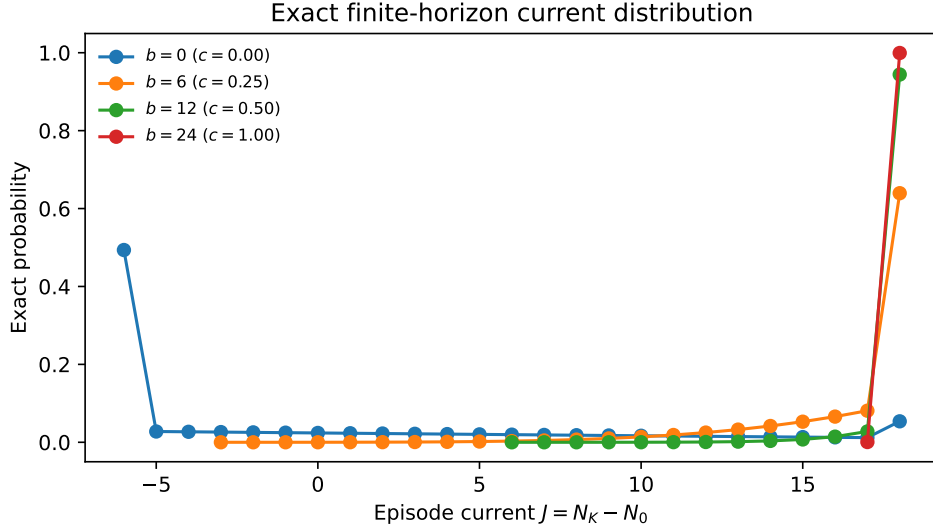


Figure 6: Exact finite-horizon current distributions for an illustrative always-advocate $q = 1$ system with $N = 24$, $K = 10$, and $n_0 = 6$. Increasing the budget moves probability mass toward larger positive currents. At strong control, the distribution also becomes much narrower because most trajectories approach the target-consensus boundary by the end of the episode.

Logic of the step

This figure is conceptually important. A mean current is only one number extracted from an entire distribution. Two controllers can have similar means but very different widths. The width is what motivates the variance and precision statistics derived next.

29 Exact finite-horizon mean and variance

We now take ordinary moments of the exact current distribution.

29.1 Mean current

For a fixed initial state n_0 ,

$$\mathbb{E}[J^{(K)} \mid n_0] = \sum_j j \Pr(J^{(K)} = j \mid n_0) \quad (148)$$

$$= \sum_m (m - n_0) [\mathcal{R}^K]_{n_0 m}. \quad (149)$$

If the initial state itself is random, average once more over P_0 :

Result

$$\mu_J = \mathbb{E}[J^{(K)}] = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0). \quad (150)$$

This is the exact theoretical counterpart of taking the arithmetic mean of the episode currents

measured across repeated simulations.

29.2 Second moment

The same calculation with J^2 gives

Result

$$M_{2,J} = \mathbb{E}[(J^{(K)})^2] = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0)^2. \quad (151)$$

29.3 Variance

Recall the elementary identity

$$\text{Var}(J) = \mathbb{E}[J^2] - \mathbb{E}[J]^2. \quad (152)$$

Substituting Eqs. (150) and (151) gives

Result

$$V_J = \text{Var}(J^{(K)}) = M_{2,J} - \mu_J^2. \quad (153)$$

Intuition

The mean answers “how much steering do I get on average?” The variance answers “if I repeat the same stochastic experiment, how much does that steering fluctuate from episode to episode?” A good controller may have a large positive mean, but if its variance is also enormous then its outcome is not very reproducible.

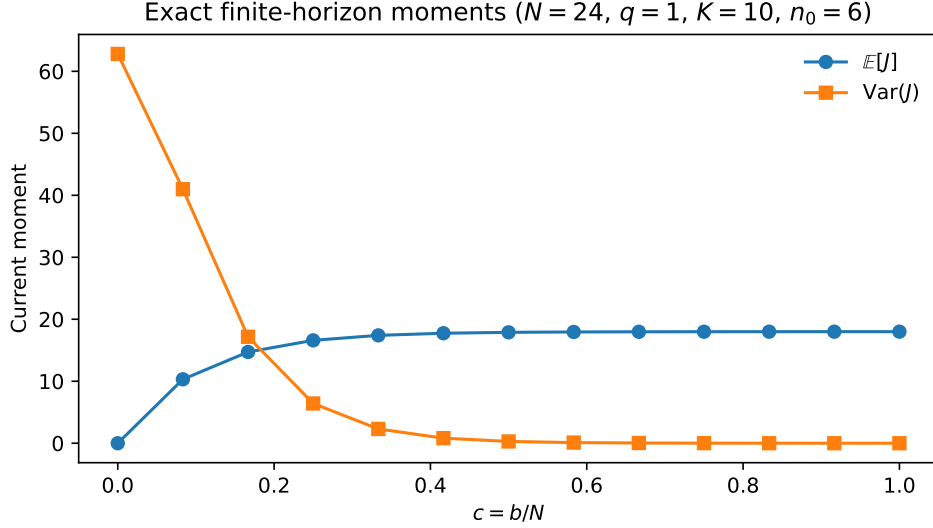


Figure 7: Exact finite-horizon mean current and variance versus actuation fraction for $N = 24$, $q = 1$, $K = 10$, and $n_0 = 6$. In this illustrative regime the mean approaches its maximum possible displacement, while the variance eventually collapses as strong control drives almost all trajectories close to target consensus.

30 Fano-like dispersion, inverse-Fano precision, and squared SNR

Mean and variance can be reported directly, but ratios of the two help summarize how large the fluctuations are relative to the directed signal.

30.1 Fano-like dispersion

Define

Result

$$F_J^{\text{disp}} = \frac{V_J}{|\mu_J|}. \quad (154)$$

The absolute value prevents the sign of the current from changing what is intended to be a fluctuation magnitude. A large F_J^{disp} means a large variance relative to the magnitude of the mean current.

Common point of confusion

The word “Fano factor” is not used consistently across literatures. A common convention is variance divided by mean magnitude, as in Eq. (154). In the stochastic-thermodynamics current-precision discussion that motivated our notation, the reciprocal ratio is emphasized. To avoid ambiguity, the report should print the formula or an explicit name rather than a bare variable called simply “Fano.”

30.2 Inverse-Fano current precision

Define the reciprocal quantity

Result

$$P_J^{\text{Iri}} = \frac{|\mu_J|}{V_J} = \frac{1}{F_J^{\text{disp}}}. \quad (155)$$

Here “Iri” is only a mnemonic for the inverse-Fano precision convention used in the social-imitation stochastic-thermodynamics discussion. In this tutorial we use it as a *statistical current-precision summary*; we are not claiming a thermodynamic uncertainty relation for the controlled model.

30.3 Squared signal-to-noise ratio

A particularly transparent dimensionless precision measure is

Result

$$\text{SNR}_J^2 = \frac{\mu_J^2}{V_J}. \quad (156)$$

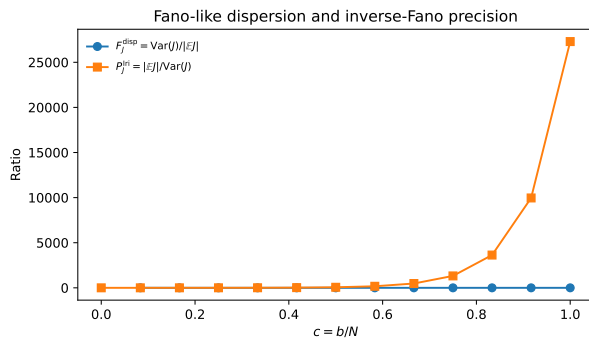
The numerator is the squared directed signal; the denominator is the stochastic fluctuation scale.

The three ratios are algebraically connected. From Eqs. (154) to (156),

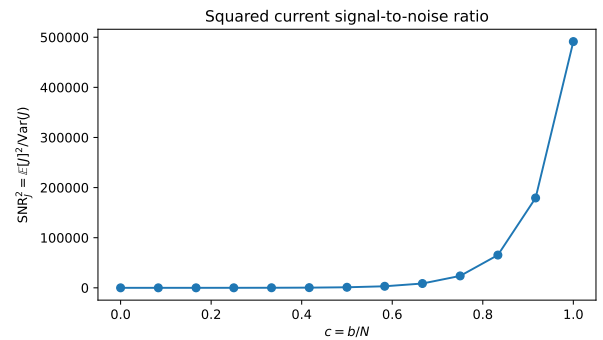
$$\text{SNR}_J^2 = |\mu_J| P_J^{\text{Iri}} \quad (157)$$

$$= \frac{|\mu_J|}{F_J^{\text{disp}}}. \quad (158)$$

So SNR^2 becomes large either because the mean displacement is large, because the current distribution is narrow, or both.



(a) Dispersion and inverse precision.



(b) Squared current SNR.

Figure 8: Exact finite-horizon fluctuation summaries for the same $N = 24$, $q = 1$, $K = 10$, $n_0 = 6$ example. The sharp increase of precision at high c is a consequence of both a mean current near its maximal value and a variance that becomes very small by the finite horizon. It should not be interpreted as a universal monotonic law for every model or horizon.

30.4 What happens when the mean or variance is zero?

The ratios need explicit conventions at degenerate points.

If $\mu_J = 0$ but $V_J > 0$, then the population fluctuates without a preferred net direction. In that case

$$\text{SNR}_J^2 = 0, \quad P_J^{\text{Iri}} = 0,$$

while $F_J^{\text{disp}} = V_J/|\mu_J|$ is undefined or divergent.

If $V_J = 0$ and $\mu_J \neq 0$, every trajectory has the same nonzero current. Then both inverse-Fano precision and SNR^2 diverge. If both are zero, the process has no current and no current fluctuations, so ratio statistics should be marked undefined rather than regularized silently with an arbitrary epsilon.

31 Worked finite-horizon example: $N = 24$, $q = 1$, $K = 10$, $n_0 = 6$

A numerical example makes the hierarchy of quantities concrete. Consider an always-Advocate controller and start from six target supporters out of twenty-four. The largest possible current is

$$J_{\max} = 24 - 6 = 18.$$

Using the exact R_1 kernel and propagating it for ten rounds gives the following values.

b	$c = b/N$	μ_J	V_J	F_J^{disp}	SNR_J^2
0	0.00	0.000	62.805	undefined	0.000
3	0.125	12.979	27.160	2.093	6.202
6	0.250	16.600	6.414	0.386	42.960
12	0.500	17.891	0.292	0.0163	1096.931
24	1.000	17.999	0.000659	3.66×10^{-5}	4.91×10^5

Logic of the step

At $b = 0$, the ordinary $q = 1$ voter is a martingale: its mean current is zero, but it still fluctuates strongly, hence the large nonzero variance. As control grows, the current becomes both more positive and more concentrated. By $b = 24$, ten rounds are long enough that almost every trajectory is extremely close to the all-target boundary, so the variance is tiny. This is why SNR^2 becomes enormous in this particular finite-horizon example.

This example also explains why mean, variance, and precision should be reported together. Looking only at the mean would miss the fact that $b = 3$ is much less reliable than $b = 12$ even though both produce substantial positive steering.

32 How empirical repeated simulations map onto the theory

Suppose the LLM experiment produces repeated episodes $e = 1, \dots, R$ for the same task and control cell. For each episode compute

$$J^{(e)} = n_Z^{(e)}(K) - n_Z^{(e)}(0), \quad (159)$$

where Z is the controller target. Then estimate

$$\hat{\mu}_J = \frac{1}{R} \sum_{e=1}^R J^{(e)}, \quad (160)$$

$$\hat{V}_J = \frac{1}{R-1} \sum_{e=1}^R (J^{(e)} - \hat{\mu}_J)^2, \quad (161)$$

and derive the empirical ratios

$$\hat{F}_J^{\text{disp}} = \frac{\hat{V}_J}{|\hat{\mu}_J|}, \quad \hat{P}_J^{\text{Iri}} = \frac{|\hat{\mu}_J|}{\hat{V}_J}, \quad \widehat{\text{SNR}}_J^2 = \frac{\hat{\mu}_J^2}{\hat{V}_J}. \quad (162)$$

The matched theory uses the same (N, q, b, K) and the same initial target-count distribution P_0 . It then evaluates Eqs. (150) and (153) to (156) exactly.

Common point of confusion

The variance here is a property of repeated stochastic trajectories. It should be estimated within a matched task/cell before pooling over heterogeneous tasks. Otherwise task-to-task differences are mixed into what is supposed to represent stochastic trajectory variability. With only two repetitions, the sample variance and all derived precision statistics are mathematically defined but statistically very unstable; they should be treated as descriptive pilot quantities.

Intuition

The theory therefore gives a direct side-by-side comparison:

$$\text{LLM: } (\hat{\mu}_J, \hat{V}_J, \hat{F}_J, \widehat{\text{SNR}}_J^2) \quad \text{versus} \quad q\text{-voter: } (\mu_J, V_J, F_J, \text{SNR}_J^2).$$

Nothing is fitted to make the two agree. The classical process is a reference response law at matched population and control resources.

33 What each reported figure is telling us

33.1 Controller-policy figure

Figure 1 answers a sensing question: after averaging over the finite sample, how often does the controller advocate at a true population fraction x ? It depends on (q_c, θ, β) but not on the population interaction order q .

33.2 Local-TE curves

Figure 2 answers a dynamical question: at a fixed current state, how distinguishable are the one-round outcomes under NoOp and Advocate?

For $q = 1$, advocacy can create target support from $x = 0$, so the boundary channel can be strong. For $q = 2$, $T(0) = 0$ and the information channel peaks in the interior, where there is enough target support to complete the controlled unanimity condition but the population is not yet saturated near $x = 1$.

33.3 Budget-scaling curve

Figure 5 answers a resource question: how much state-local information can the controller inject as its actuation fraction $c = b/N$ grows? For $q \geq 2$, the initial growth follows the c^2 prediction of the weak-control expansion before finite-separation effects become important.

33.4 Closed-form $q = 1$ current figure

Figure 4 is a consistency figure. The solid line is produced by the full exact whole-round kernel R_1 ; the dashed line is the scalar closed-form expression. Their agreement demonstrates that the current observable is genuinely derived from the same population kernel rather than introduced as an unrelated empirical metric.

33.5 Current-distribution and precision figures

Figure 6 shows the complete finite-horizon distribution before it is compressed into moments. Figure 7 then extracts the first two moments, while Fig. 8 forms relative fluctuation summaries from those moments. Reading them in that order is useful: distribution $\rightarrow$ mean/variance $\rightarrow$ precision ratios.

34 A compact computational recipe

For a specified parameter tuple

$$(N, q, q_c, b, \beta, \theta),$$

the exact theoretical calculation is:

- Step 1.** For each n , compute the sensor distribution $S(y | n)$ from Eq. (24).
- Step 2.** Average the sigmoid policy over sensor outcomes to obtain a_n from Eq. (28).
- Step 3.** Build the ordinary one-update kernel K_0 from Eqs. (8) to (10).
- Step 4.** Build the controlled one-update kernel K_1 from Eqs. (11) and (12).
- Step 5.** Compute the NoOp round kernel $R_0 = K_0^N$.

- Step 6.** Compute the advocacy round kernel $R_1 = F_{N,b}$ using the exact recursion Eqs. (53) and (54).
- Step 7.** At each state n , form the action-marginalized next-state law M_n from Eq. (61).
- Step 8.** Evaluate the weighted Jensen–Shannon divergence $T(n)$ using Eq. (65) or Eq. (68).
- Step 9.** If an initial distribution P_0 is specified, propagate it using Eq. (71) and average $T(n)$ to obtain T_k and $T^{(K)}$.
- Step 10.** For a finite-horizon current experiment choose $\mathcal{R} = R_0, R_1$, or the feedback mixture R according to the controller mode.
- Step 11.** Propagate K rounds with $\mathcal{R}^K$ and obtain the exact current distribution from Eqs. (146) and (147).
- Step 12.** Compute the exact current mean and variance from Eqs. (150) and (153).
- Step 13.** Compute F_J^{disp} , P_J^{Iri} , and SNR_J^2 from the same two moments, keeping zero-mean and zero-variance cases explicit.

35 Sanity checks that are useful when implementing the theory

A correct numerical implementation should satisfy the following checks.

35.1 Every microscopic kernel row sums to one

For every n ,

$$\sum_m K_0(m | n) = 1, \quad \sum_m K_1(m | n) = 1.$$

35.2 Every whole-round kernel row sums to one

For every n ,

$$\sum_m R_0(m | n) = 1, \quad \sum_m R_1(m | n) = 1.$$

35.3 The sensor law sums to one

For every n ,

$$\sum_y S(y | n) = 1.$$

35.4 Advocacy probability is valid

$$0 \leq a_n \leq 1.$$

For $q_c = 2$, $\theta = 1/2$, verify numerically that Eq. (28) agrees with the closed form Eq. (44).

35.5 Transfer entropy is nonnegative

Up to floating-point roundoff,

$$T(n) \geq 0.$$

Any tiny negative values of order 10^{-15} are numerical noise and should be interpreted as zero.

35.6 Boundary checks

For $q \geq 2$,

$$T(0) = 0.$$

For $q = 1$ and $b > 0$, the exact theory should give

$$T(0) > 0$$

whenever $0 < a_0 < 1$.

35.7 Current telescoping check

When microscopic trajectories are available,

$$\sum_t j_t = N_K - N_0. \quad (163)$$

This should hold exactly episode by episode.

35.8 $q = 1$ closed-form current check

For $q = 1$, the mean current computed from R_1 must agree with [Eq. \(123\)](#) for every n and b up to numerical precision.

35.9 Finite-horizon distribution normalizes

For fixed n_0 ,

$$\sum_m [\mathcal{R}^K]_{n_0 m} = 1, \quad (164)$$

and equivalently

$$\sum_j \Pr(J = j \mid n_0) = 1. \quad (165)$$

35.10 Variance is nonnegative

Numerically, V_J from [Eq. \(153\)](#) should satisfy $V_J \geq 0$ up to tiny floating-point error. A small value such as -10^{-14} can be clipped to zero only after confirming it is numerical roundoff rather than a derivation or implementation error.

36 How to interpret the theory as a benchmark for multi-agent LLM experiments

The classical calculation separates three resources that can otherwise become entangled in a language-agent experiment:

$$\begin{aligned} q_c & : \text{sensing resource,} \\ (\beta, \theta) & : \text{controller decision policy,} \\ c = b/N & : \text{actuation resource,} \\ q & : \text{population response nonlinearity.} \end{aligned}$$

At matched parameter values, the classical model gives an exact reference channel

$$T_{\text{classical}}(n)$$

without sampling error in the mutual-information calculation.

The comparison with a multi-agent LLM system should therefore ask not only whether the empirical TE is large or small, but *where the discrepancy enters*:

- Is the LLM controller’s effective action variation comparable to $a_n(1 - a_n)$?
- Does advocacy actually change the next-state law, i.e. is there an empirical analogue of $R_0 \neq R_1$?
- Does the effect scale with the actuation budget b as the classical model predicts?
- Does increasing the social interaction order q suppress boundary controllability in the expected way?

Intuition

The exact theory is useful because it turns “control works” into two separately testable statements: the controller must vary its action, and the population must react differently to those actions. Transfer entropy becomes large only when both are true.

The finite-horizon current observables provide a second comparison axis. Transfer entropy compares the *information channel*; current compares the *behavioral transport*. In particular, open-loop always-Advocate experiments have zero action TE because U_k is deterministic, but they still have a meaningful and often nonzero current distribution. Those experiments should therefore be interpreted as response/susceptibility or current experiments rather than TE experiments.

For relational LLM populations with three semantic answers, the binary classical theory should be matched by coarse-graining to

$$\text{controller target } Z \quad \text{versus} \quad \text{not } Z.$$

The empirical truth current can still be reported separately, but when the controller target is adversarial it is not the same observable as the matched target current.

37 Relation to the theoretical anchors cited in the source note

The source theory note points to two broader theoretical anchors.

Irisarri et al. develop a stochastic-thermodynamic framework for nonlinear social imitation, motivating the use of nonlinear q -dependent imitation dynamics and information/flux language in social systems. Their framework emphasizes microscopic reversibility for their thermodynamic construction, whereas the controlled kernel used here is intentionally asymmetric: at a controlled position, the controller message supporting A makes the reverse $A \rightarrow B$ move impossible. The finite- N transfer-entropy calculation developed in this tutorial does not require microscopic reversibility.

Horowitz and Sandberg analyze information measures in repeated feedback systems and motivate transfer entropy as a directional information quantity in feedback. In the present discrete round-level setting, the corresponding channel is precisely

$$I(U_k; N_{k+1} \mid N_k).$$

The current report uses that directional-information idea without importing a thermodynamic work/heat interpretation.

The current extension also has a useful conceptual connection to the social-imitation stochastic-thermodynamics literature, where a trajectory current is defined as a signed excess of forward over reverse opinion-change events and its fluctuations are used to discuss current precision. Our $J = N_K - N_0$ is the corresponding finite-horizon net target displacement for the controlled process. We deliberately stop at the statistical observables—mean, variance, Fano-like dispersion, inverse-Fano precision, and SNR^2 —because the controlled kernel contains strictly one-sided transitions and we have not derived a thermodynamic uncertainty relation for it.

38 One-page summary of the derivation

From model definition to exact transfer entropy

1. Ordinary microscopic response

$$K_0(n+1 \mid n) = \frac{N-n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}}, \quad K_0(n-1 \mid n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}}.$$

2. Controlled microscopic response

$$K_1(n+1 \mid n) = \frac{N-n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}}, \quad K_1(n-1 \mid n) = 0.$$

3. Finite sensor

$$S(y \mid n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}.$$

4. Soft controller averaged over sensing noise

$$a_n = \sum_y S(y | n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right].$$

For $q_c = 2$, $\theta = 1/2$,

$$a(x) = \frac{1}{2} + \frac{1}{2} \tanh \left(\frac{\beta}{4} \right) (1 - 2x).$$

5. Whole-round dynamics

$$R_0 = K_0^N, \quad R_1 = F_{N,b},$$

with

$$F_{r+1,j+1} = \frac{b-j}{N-r} F_{r,j} K_1, \quad F_{r+1,j} = \frac{N-b-r+j}{N-r} F_{r,j} K_0.$$

6. Marginalize the controller action

$$M_n = (1 - a_n) R_0 + a_n R_1.$$

7. Exact state-local transfer entropy

$$T(n) = \text{JS}_{a_n}(R_0(\cdot | n), R_1(\cdot | n)).$$

Equivalently,

$$T(n) = (1 - a_n) D_{\text{KL}}(R_0 \| M_n) + a_n D_{\text{KL}}(R_1 \| M_n).$$

8. Large- N weak-control approximation

$$f_0 = (1 - x)x^q - x(1 - x)^q, \quad f_1 = (1 - x)x^{q-1},$$

$$\Delta f_q = x^{q-1}(1 - x)^2 + x(1 - x)^q,$$

$$\nu_0 = (1 - x)x^q + x(1 - x)^q - f_0^2,$$

$$T_{\text{MF}}(x) \approx \frac{a(1-a)}{2 \ln 2} N c^2 \frac{\Delta f_q^2}{\nu_0}.$$

9. Special boundary result

$$q = 1 : T(0) > 0 \text{ for } b > 0, \quad q \geq 2 : T(0) = 0.$$

10. Finite-horizon target current

$$J^{(K)} = N_K - N_0, \quad \Pr(N_K = m | N_0 = n_0) = [\mathcal{R}^K]_{n_0 m}.$$

11. Exact current mean and variance

$$\mu_J = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0),$$

$$V_J = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0)^2 - \mu_J^2.$$

12. Current fluctuation summaries

$$F_J^{\text{disp}} = \frac{V_J}{|\mu_J|}, \quad P_J^{\text{Iri}} = \frac{|\mu_J|}{V_J}, \quad \text{SNR}_J^2 = \frac{\mu_J^2}{V_J}.$$

13. Closed-form $q = 1$ one-round mean current

$$\mu_{J,1}^{(q=1)}(n) = (N - n) \left[1 - \left(1 - \frac{1}{N} \right)^b \right].$$

39 Final conceptual picture

The entire calculation can be read as the construction of three ingredients:

$$\boxed{a_n}, \quad \boxed{R_0(\cdot \mid n)}, \quad \boxed{R_1(\cdot \mid n)}.$$

The first tells us whether the controller sometimes advocates and sometimes does not. The latter two tell us whether those actions actually generate different population futures.

The exact theory then says

$$\boxed{T(n) = \text{JS}_{a_n}(R_0, R_1)}.$$

The weak-control approximation says the same thing in signal-to-noise language:

$$T_{\text{MF}} \approx \underbrace{a(1-a)}_{\text{action variability}} \underbrace{\frac{Nc^2}{2 \ln 2}}_{\text{actuation scale}} \underbrace{\frac{\Delta f_q^2}{\nu_0}}_{\text{response-to-noise ratio}}.$$

Thus the core lesson is simple:

Control information is large when the controller has action variability, its action changes the population dynamics strongly, and that change is large compared with the population's intrinsic stochastic fluctuations.

The current calculation adds a second summary of the same kernels:

$$\boxed{\mathcal{R}^K \implies \text{Pr}(J) \implies (\mu_J, V_J) \implies (F_J^{\text{disp}}, P_J^{\text{Iri}}, \text{SNR}_J^2)}.$$

The two families should be read together:

Transfer entropy tells us how much controller-action information is expressed in the population transition law. Current tells us how far the population actually moves and how reproducible that motion is. The same exact finite- N kernels generate both.

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